

Weijie Du

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EDUCATION

Carnegie Mellon University — Master of Science in Software Engineering Jan 2025 – May 2026
Relevant Courses: Foundations of Software Engineering, Data Science, Interaction Design, Mobile Embedded Software

University of California, Irvine — Bachelor of Science in Computer Science Sep 2020 – Jun 2024
Relevant Courses: Algorithms & Data Structures, Machine Learning, Deep Learning, Data Management, Software Testing

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C++, HTML/CSS, Bash, F#

Frameworks & Libraries: React, Next.js, Node.js, Express, Redux, Bootstrap, TensorFlow, PyTorch, CLIP, LSeg

Tools: Docker, Git, Firebase, PostgreSQL, MongoDB, MySQL, Socket.IO, Render, Netlify, OpenAI API, NumPy, OpenCV, NLTK

EXPERIENCE

BOSCH (School Collaboration) Mountain View, CA
Software Engineer Intern Aug 2025 – Dec 2025

- Improved autonomous parking precision by achieving **centimeter-level localization accuracy** through engineering a **semantic 3D mapping pipeline**, aligned closely with research scientists' technical direction
- Reduced manual annotation effort by integrating **Vision-Language Models (CLIP, LSeg)** with geometric **point-cloud algorithms** to automate detection of lanes, curbs, and road markings, guided by research-driven requirements
- Accelerated map load times by **80%** by optimizing map asset storage through a compression pipeline that cut data size from **700 MB** to **130 MB**, developed in consultation with the research scientist lead
- Improved developer iteration speed by **reducing asset load latency by 99%** through targeted **caching and warm-start optimizations** for interactive 3D visualization, informed by research-backed performance goals

FACT FINDERS PRO Seattle, WA
Software Engineer Intern Oct 2024 – Jan 2025

- Enhanced usability for non-technical audiences by developing an **AI-powered fact-checking platform** with **React + TypeScript**
- Enabled faster verification throughput by implementing **Python REST APIs** for real-time data retrieval and analysis
- Ensured smooth design-to-engineering handoffs by collaborating with cross-functional teams to translate product requirements into polished, scalable features

UNIVERSITY OF CALIFORNIA, IRVINE Irvine, CA
Research Assistant Sep 2022 – Jun 2024

- Co-authored two conference publications at **PACMHCI '25** and **ASSETS '24** by collecting, coding, and analyzing **200+ qualitative data points** on noise sensitivity and social support patterns in autistic communities.
- Led over **10 interviews and co-design sessions** with individuals with autism, generating inclusive design insights that informed mobile and wearable prototype development for sensory management.
- Collaborated on a **Samsung Galaxy Watch** research prototype using **Android APIs** and the **Samsung Health SDK**, supporting studies on sensory-aware interventions in daily contexts.

PROJECTS

MindTrack [<https://github.com/weijiedu/MindTrack>] June 2025 – July 2025
React, Express, MongoDB, OpenAI API, Render

- Engineered a **MERN** journaling app enabling users to log emotions and visualize mood patterns with interactive charts
- Integrated the **OpenAI API** to analyze journal entries, providing personalized AI reflections that improved daily engagement
- Deployed with **JWT authentication** via **Render + Netlify**, achieving responsive cross-platform performance

Emergency Social Network [<https://github.com/weijiedu/emergency-social-network>] Jan 2025 – May 2025
Bootstrap, Node.js, Express, Socket.IO, PostgreSQL

- Developed a disaster-response platform allowing real-time communication using **Node.js/Express** and **PostgreSQL**
- Implemented low-latency, bi-directional messaging with **Socket.IO**, enabling instant alerts and private communications
- Improved reliability by writing **190+ automated tests (71% coverage)** and creating a **CI/CD pipeline** under an **Agile** workflow

AquaZenith Health [<https://github.com/weijiedu/AquaZenith.Health>] Jan 2024 – Mar 2024
React Native, ChatGPT API, Firebase, Apple HealthKit

- Built a wellness app with **React Native**, integrating **Apple HealthKit** and **Firebase** to monitor health data live
- Developed an **AI-powered interface** using the **ChatGPT API** to deliver personalized health insights without manual tracking
- Streamlined backend synchronization and optimized mobile UX, increasing active engagement